

Lecture #15, Nov 04, 2025

Conservation law and source of instability

Goal: determine dynamic modes of hyperbolic system

HW3 can be treated as samples for final.

Submit ans. one question on one page

Show steps. No need to type in

From Lect #WS

$$u(x,t) \in \mathbb{R}^n$$

$u(x,t)$ is vector

$$\frac{\partial u}{\partial t} + \frac{df(u)}{dx} = 0$$

$$\frac{\partial u}{\partial t} + A \frac{\partial u}{\partial x} = 0, \quad A = \frac{\partial f}{\partial u} = f'(u)$$

$$u(x,0) = \bar{u}(x)$$

A is real, distinct eigenvalues

$$A = \phi \Lambda \phi^{-1}$$

$$q(x,t) = \phi^{-1} u(x,t)$$

$$\frac{\partial q_k}{\partial t} + \lambda_k \frac{\partial q_k}{\partial x} = 0,$$

$$q = \begin{bmatrix} q_1 \\ q_2 \\ \vdots \\ q_k \\ \vdots \\ q_n \end{bmatrix}$$

Let ψ_k^T be k -th row of ϕ^{-1}

ϕ_k be k -th col of ϕ

$$q_k(x,t) = \psi_k^T u_0(x - \lambda_k t)$$

$$u(x,t) = \phi q(x,t) = \sum_{k=1}^n \psi_k^T u_0(x - \lambda_k t) \phi_k$$



Ex: $A = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$ $u \in \mathbb{R}^2$, $\frac{\partial u}{\partial t} + A \frac{\partial u}{\partial x} = 0$ ($u_t + A u_x = 0$)

Want to find $\otimes$

$$u = \begin{bmatrix} v \\ w \end{bmatrix}, \begin{cases} \frac{\partial v}{\partial t} + \frac{\partial v}{\partial x} + \frac{\partial w}{\partial x} = 0 \\ \frac{\partial w}{\partial t} + \frac{\partial v}{\partial x} + \frac{\partial w}{\partial x} = 0 \end{cases}$$

$$\Rightarrow \frac{d}{dt} \begin{bmatrix} v \\ w \end{bmatrix} + \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} \frac{\partial v}{\partial x} \\ \frac{\partial w}{\partial x} \end{bmatrix} = 0$$

eig. val/vec of A

$$\begin{vmatrix} 1-\lambda & 1 \\ 1 & 1-\lambda \end{vmatrix} = 0 \Rightarrow \lambda = 0, 2$$

$$\lambda_1 = 0, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow x = 1, y = -1, \phi_1 = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\lambda_2 = 2, \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 2 \begin{pmatrix} x \\ y \end{pmatrix} \Rightarrow x = 1, y = 1, \phi_2 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

Then $\phi = \begin{bmatrix} 1 & 1 \\ -1 & 1 \end{bmatrix}$ $\Lambda = \begin{bmatrix} 0 & 0 \\ 0 & 2 \end{bmatrix}$, $A = \phi \Lambda \phi^{-1}$

$$A\phi = \phi\Lambda$$

ϕ_1, ϕ_2 are right eigenvectors

$$\phi^{-1} = \frac{1}{2} \begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \psi^T$$

$$\psi_1^T = \frac{1}{2} (1, -1), \psi_2^T = \frac{1}{2} (1, 1)$$

(an check: $\psi^T \phi = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ (H.W))

$$u(x, t) = \sum_{k=1}^2 \psi_k^T u_0(x - \lambda_k t) \phi_k$$

$$= \psi_1^T u_0(x) \phi_1 + \psi_2^T u_0(x-2t) \phi_2$$

Ex: Repeat: $u(x, 0) = u_0(x) = \begin{bmatrix} \sin x \\ \cos x \end{bmatrix}$

$$\psi_1^T u_0(x) = \frac{1}{2} (1, -1) \begin{bmatrix} \sin x \\ \cos x \end{bmatrix} = \frac{1}{2} (\sin x - \cos x)$$

$$\psi_2^T u_0(x) = \frac{1}{2} (1, +1) \begin{bmatrix} \sin x \\ \cos x \end{bmatrix} = \frac{1}{2} (\sin x + \cos x)$$

$$q_1(x, t) = \psi_1^T u_0(x - \lambda_1 t) = \frac{1}{2} (\sin x - \cos x), \lambda_1 = 0$$

$$q_2(x, t) = \psi_2^T u_0(x - \lambda_2 t) = \frac{1}{2} \sin(x - 2t) + \frac{1}{2} \cos(x - 2t), \lambda_2 = 2$$

$$u(x, t) = q_1 \phi_1 + q_2 \phi_2 = q_1 \begin{pmatrix} 1 \\ -1 \end{pmatrix} + q_2 \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$= \begin{pmatrix} \frac{1}{2} (\sin x - \cos x) + \frac{1}{2} \sin(x - 2t) + \frac{1}{2} \cos(x - 2t) \\ -\frac{1}{2} (\sin x - \cos x) + \frac{1}{2} \sin(x - 2t) + \frac{1}{2} \cos(x - 2t) \end{pmatrix}$$

Ex: $u_t + A u_x = 0, \quad A = \begin{pmatrix} 2 & 1 \\ 0 & 1 \end{pmatrix}$

$$u(x, 0) = u_0(x) = \begin{bmatrix} \sin x \\ e^{-x^2} \end{bmatrix}$$

Find $u(x, t)$

Ex: Flux form $\left\{ \begin{array}{l} \frac{\partial P}{\partial t} + \frac{\partial}{\partial x}(K_0 u) = 0 \\ \rho_0 \frac{\partial u}{\partial t} + \frac{\partial P}{\partial x} = 0 \end{array} \right. \Rightarrow \frac{\partial u}{\partial t} + \frac{\partial}{\partial x} \left(\frac{1}{\rho_0} P \right) = 0$

Matrix form: $\left\{ \begin{array}{l} \frac{\partial P}{\partial t} + K_0 \frac{\partial u}{\partial x} = 0 \\ \frac{\partial u}{\partial t} + \frac{1}{\rho_0} \frac{\partial P}{\partial x} = 0 \end{array} \right. \quad A = \begin{bmatrix} 0 & K_0 \\ \frac{1}{\rho_0} & 0 \end{bmatrix}$
 $\vec{u} = \begin{bmatrix} P \\ u \end{bmatrix}$

eigenvalues: $|A - \lambda I| = 0$

$\lambda_1 = -c_0, \lambda_2 = c_0, \quad c_0 = \frac{K_0}{\rho_0}$

eigenvectors $\phi_1 = \begin{bmatrix} -z_0 \\ 1 \end{bmatrix}, \phi_2 = \begin{bmatrix} z_0 \\ 1 \end{bmatrix}, \quad z_0 = \rho_0 c_0$

$\phi = \begin{bmatrix} -z_0 & z_0 \\ 1 & 1 \end{bmatrix}$

$\phi^{-1} = \frac{1}{2z_0} \begin{bmatrix} -1 & z_0 \\ 1 & z_0 \end{bmatrix} \begin{matrix} \leftarrow \psi_1^T \\ \leftarrow \psi_2^T \end{matrix}$

$\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} \right)^{-1} = \frac{1}{\det} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$

$q_1 = \psi_1^T u_0(x) = \frac{1}{2z_0} [-P_0(x) + z_0 u_0(x)]$

$q_2 = \psi_2^T u_0(x) = \dots$

Now, show or derive

$$p(x,t) = \frac{1}{2} [P_0(x+c_0 t) + P_0(x-c_0 t)] \\ - \frac{z_0}{2} [u_0(x+c_0 t) - u_0(x-c_0 t)]$$

$$u(x,t) = - \frac{1}{2z_0} [P_0(x+c_0 t) - P_0(x-c_0 t)] \\ + \frac{1}{2} [u_0(x+c_0 t) + u_0(x-c_0 t)]$$

H.W. $u(x,t) = \begin{bmatrix} v \\ w \end{bmatrix} \cdot A = \begin{bmatrix} 0 & -c^2 \\ -1 & 0 \end{bmatrix}$

$$u(x,0) = \begin{cases} u_L & x < 0 \\ u_R & x > 0 \end{cases}, \quad u_L \text{ and } u_R \text{ are vectors}$$

Find an expression for upwind flux $F(u_L, u_R)$
at $x_{j+\frac{1}{2}}$

(To be covered in next lectures)

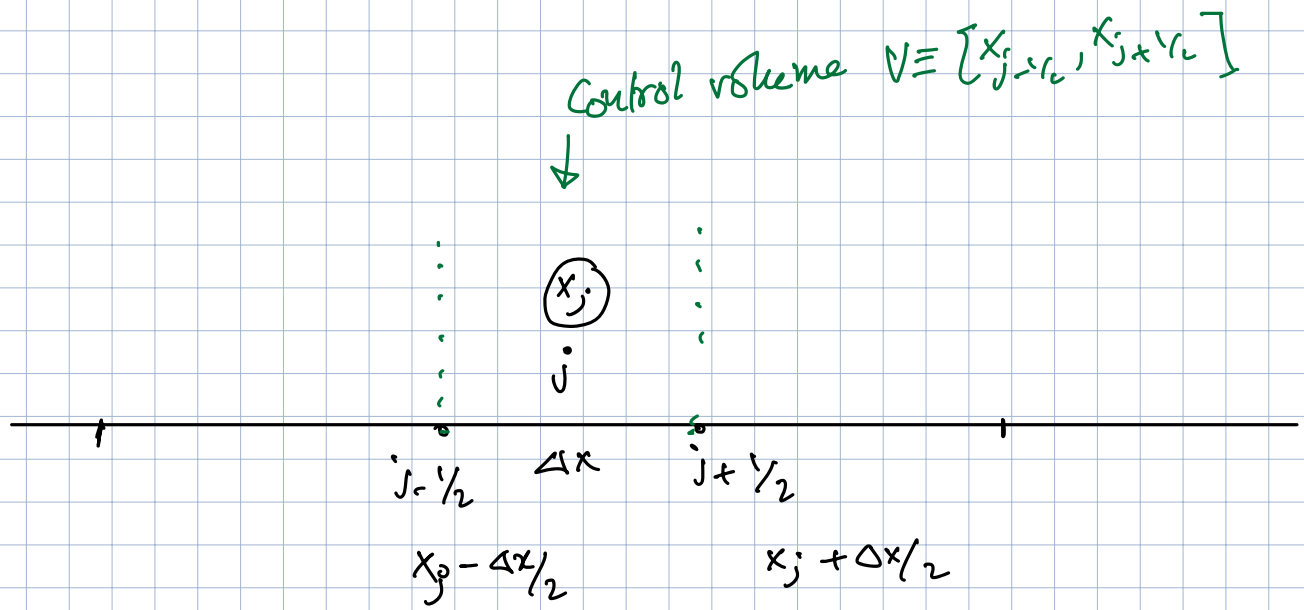
Conservation law (see W#8 lect)

$$\frac{d}{dt} \int_V u \, dx + \int_{\partial V} f(u) \cdot \hat{n} \, dS = \int_V s(u) \, dx$$

change of solution
in control volume.

inflow/outflow
in V

source



Here Δx not same for all C.V.

let $\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0$, $f(u)$ is flux

let $u_j(t) \approx \frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} u(x, t) dx$ (cell value in F.V.M)

Integrate $\int_{x_{j-1/2}}^{x_{j+1/2}} \frac{\partial u}{\partial t} dx + f(u(x_{j+1/2}, t)) - f(u(x_{j-1/2}, t)) = 0$

$\div$ by Δx $\frac{\partial u_j(t)}{\partial t} + \frac{1}{\Delta x} [f_{j+1/2} - f_{j-1/2}] = 0$

$$\frac{\partial u_j}{\partial t} = - \frac{1}{\Delta x} [f_{j+1/2} - f_{j-1/2}]$$

Note. $f_{j+1/2}$ is flux value at right face of C.V.
 $f_{j-1/2}$ " " " " left " C.V.

Solution is stored at j -th C.V or (j -th cell)

Hence flux values are not known.

In FVM, we have to estimate flux values.

flux at $x_{j \pm 1/2}$ will be estimated using solution at right and left c.v.s

$$\text{Simple case} \Rightarrow f_{j+1/2} = F(u_j, u_{j+1})$$

$$f_{j-1/2} = F(u_{j-1}, u_j)$$

using this

$$\frac{du_j(t)}{dt} = - \frac{1}{\Delta x} [F(u_j, u_{j+1}) - F(u_{j-1}, u_j)]$$

↑
numerical flux function

$$\text{let } F(u_j, u_{j+1}) = \frac{1}{2} f(u_j) + \frac{1}{2} f(u_{j+1})$$

$$F(u_{j-1}, u_j) = \frac{1}{2} f(u_{j-1}) + \frac{1}{2} f(u_j)$$

$$\frac{du_j}{dt} = - \frac{1}{2\Delta x} [f(u_{j+1}) - f(u_{j-1})]$$

discretize in time.

$$u_j^{n+1} = u_j^n - \frac{\Delta t}{2\Delta x} [f(u_{j+1}) - f(u_{j-1})]$$

Center approximation

Lecture #16, Nov 06, 2025

small hint: HW3

Q2(e)

$$\begin{bmatrix} p_e \\ u_e \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \quad \text{if } x < 0$$

$$\begin{bmatrix} p_r \\ u_r \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \quad \text{if } x > 0$$

Q2f

Convert the system to

$$\frac{d}{dt} \begin{bmatrix} \rho \\ u \end{bmatrix} + A \begin{bmatrix} \frac{\partial \rho}{\partial x} \\ \frac{\partial u}{\partial x} \end{bmatrix} = 0$$

To remove p from the system

$$\frac{\partial p}{\partial x} = \frac{\partial p}{\partial \rho} \frac{\partial \rho}{\partial x} \quad \text{use } a^v = \frac{dp}{d\rho}$$

Final exam will have distinct script for 6201 and 4162

Ex: For $\frac{du}{dt} + \frac{d}{dx} f(u) = 0$

derive the finite volume discretization

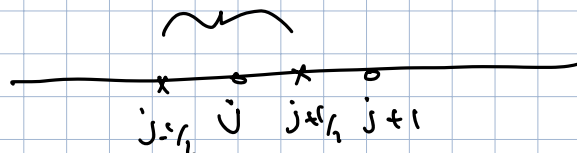
$$u_j^{n+1} = u_j^n - \frac{\Delta t}{\Delta x} (F_{j+1/2}^n - F_{j-1/2}^n)$$

and define the variables e.g. grid function u_j^n

and flux function $F_{j-1/2}^n$

Sol:

Take



$$\int_{x_{j-1/2}}^{x_{j+1/2}} \frac{du}{dt} dx + f(u(x_{j+1/2}, t)) - f(u(x_{j-1/2}, t)) = 0$$

$$\frac{d}{dt} u_j + \frac{1}{\Delta x} [f_{j+1/2} - f_{j-1/2}] = 0$$

$$u_j(t) = \frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} u(x, t) dx$$

$$\int_{t_n}^{t_{n+1}} \frac{du_j}{dt} dt + \frac{1}{\Delta x} \left[\int_{t_n}^{t_{n+1}} f_{j+1/2} dt - \int_{t_n}^{t_{n+1}} f_{j-1/2} dt \right] = 0$$

$$u_j(t_{n+1}) - u_j(t_n) + \frac{\Delta t}{\Delta x} \left[\frac{1}{\Delta t} \int_{t_n}^{t_{n+1}} f_{j+1/2} dt - \dots \right] = 0$$

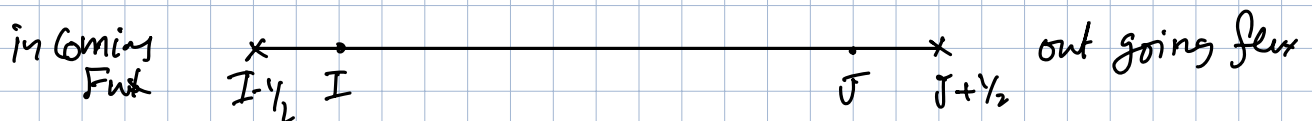
$$F_{j+1/2} = \frac{1}{\Delta t} \int_{t_n}^{t_{n+1}} f(u(x_{j+1/2}, t)) dt$$

$$\Rightarrow \boxed{u_j^{n+1} = u_j^n - \frac{\Delta t}{\Delta x} (F_{j+1/2}^n - F_{j-1/2}^n)} \quad \begin{array}{l} \text{standard FVM} \\ \text{Euler explicit FVM} \end{array}$$

Ex: On a grid $\{x_j\}_{j=I}^{j=J}$

show that

$$\sum_{j=I}^J u_j^{n+1} = \sum_{j=I}^J u_j^n - \frac{\Delta t}{\Delta x} (F_{J+1/2}^n - F_{I-1/2}^n)$$



Hint:

$$u_j^{n+1} = u_j^n - \frac{\Delta t}{\Delta x} (F_{j+1/2}^n - F_{j-1/2}^n)$$

$$\begin{aligned} \sum_{j=I}^J () &= \sum_{j=I}^J () - \frac{\Delta t}{\Delta x} \left[\cancel{F_{I+1/2}^n} - F_{I-1/2}^n \right. \\ &\quad \left. + F_{I+3/2}^n - \cancel{F_{I+1/2}^n} + \dots \dots \dots F_{J+1/2}^n - \cancel{()} \right] \end{aligned}$$

How to deal with parabolic eqⁿ with variable or discontinuous coefficients

Ex: Derive FV scheme for

$$\frac{\partial u}{\partial t} = \frac{\partial}{\partial x} \left(v(x) \frac{\partial u}{\partial x} \right)$$

Solⁿ

Begin with

$$\frac{\partial u}{\partial t} + \frac{\partial f(u)}{\partial x} = 0$$

Flux form

$$\frac{\partial u}{\partial t} + \nabla \cdot (\vec{a} u) = 0$$

$$\int_V \frac{\partial u}{\partial t} dV + \int_V \nabla \cdot \vec{a} u dV = 0$$

$$\frac{\partial}{\partial t} \int_V u dV + \underbrace{\int_S \vec{a} u \cdot \hat{n} dA}_{\text{flux}} = 0$$

For parabolic eqⁿ $f(u) = -v(x) \frac{\partial u}{\partial x}$

Integrate over a cell $[x_{j-1/2}, x_{j+1/2}]$ and $\div$ by Δx

$$\frac{d}{dt} \left[\frac{1}{\Delta x} \int_{x_{j-1/2}}^{x_{j+1/2}} u(x, t) dx + \frac{1}{\Delta x} \left[-v_{j+1/2} \frac{\partial u}{\partial x} \Big|_{j+1/2} + v_{j-1/2} \frac{\partial u}{\partial x} \Big|_{j-1/2} \right] \right] = 0$$

↑ note this v_{xx} factor.

(*)

$$\frac{\partial u_j(t)}{\partial t} = \frac{1}{\Delta x} \left[v_{j+1/2} \frac{\partial u}{\partial x} \Big|_{j+1/2} - v_{j-1/2} \frac{\partial u}{\partial x} \Big|_{j-1/2} \right]$$

FVM defines flux function as

$$\begin{aligned} F_{j-1/2}^n &= \frac{1}{\Delta t} \int_{t_n}^{t_{n+1}} f(u(x_{j-1/2}, t)) dt \\ &= \frac{1}{\Delta t} \int_{t_n}^{t_{n+1}} -v_{j-1/2} \frac{\partial u}{\partial x} \Big|_{j-1/2} dt \end{aligned}$$

Integrate (*) over $[t_n, t_{n+1}]$

$$\int_{t_n}^{t_{n+1}} \frac{d}{dt} u_j(t) dt = \frac{\Delta t}{\Delta x} \left[\int_{t_n}^{t_{n+1}} v_{j+1/2} \frac{\partial u}{\partial x} \Big|_{j+1/2} dt - \int_{t_n}^{t_{n+1}} v_{j-1/2} \frac{\partial u}{\partial x} \Big|_{j-1/2} dt \right]$$

$$u_j(t_{n+1}) - u_j(t_n) = \frac{\Delta t}{\Delta x} \left[v_{j+1/2} \frac{u_{j+1}^n - u_j^n}{\Delta x} - v_{j-1/2} \frac{u_j^n - u_{j-1}^n}{\Delta x} \right]$$

Here $\frac{1}{\Delta t} \int_{t_n}^{t_{n+1}} v_{j+1/2} \frac{\partial u}{\partial x} \bigg|_{j+1/2} dt = v_{j+1/2} \frac{\partial u}{\partial x} \bigg|_{j+1/2}^n$

$v(x)$ does not depend on t

$\frac{\partial u}{\partial x}$ has been evaluated at t_n

$$u_j^{n+1} = u_j^n + \frac{\Delta t}{\Delta x^2} \left[v_{j+1/2} (u_{j+1}^n - u_j^n) - v_{j-1/2} (u_j^n - u_{j-1}^n) \right]$$

Remark:

Constant

If $v(x) \equiv v$

$$\begin{aligned} u_j^{n+1} &= u_j^n - \frac{\Delta t}{\Delta x^2} (v(u_{j+1}^n - u_j^n) - v(u_j^n - u_{j-1}^n)) \\ &= u_j^n - \frac{v\Delta t}{\Delta x^2} (u_{j+1}^n - 2u_j^n + u_{j-1}^n) \quad (\text{FTCS}) \end{aligned}$$

Remark: FTCS scheme for heat eqⁿ is conditionally stable.

$$\frac{v\Delta t}{\Delta x^2} \leq \frac{1}{2}$$

$$\Delta t = O\left(\frac{\Delta x^2}{v}\right)$$

Crank-Nicolson (C-N)

define flux function

$$F_{j-1/2}^n = \frac{1}{2} [f(u(x_{j-1/2}, t_n) + f(u(x_{j-1/2}, t_{n+1}))]$$

this will give for parabolic problem

$$u_j^{n+1} = u_j^n + \frac{\Delta t}{2\Delta x^2} [v_{j+1/2}^n (u_{j+1}^n - u_j^n) - v_{j-1/2}^n (u_j^n - u_{j-1}^n)$$

$$v_{j+1/2}^{n+1} (u_{j+1}^{n+1} - u_j^{n+1}) - v_{j-1/2}^{n+1} (u_j^{n+1} - u_{j-1}^{n+1})]$$

Remark.

- time integration can be Euler explicit, Euler implicit or trapezoidal type.

- Dealing with flux is a critical aspect of FVM.
